Do not write outside the box

0 1 Alkanes and alkenes are hydrocarbons.

0 1 . 1 Define the term 'hydrocarbon'.

[1 mark]

_____

_____

0 1 . 2 The general formula for alkanes is C_nH_{2n+2}

Determine the formula of the alkane with 10 carbon atoms.

[1 mark]

_____

Formula = _____

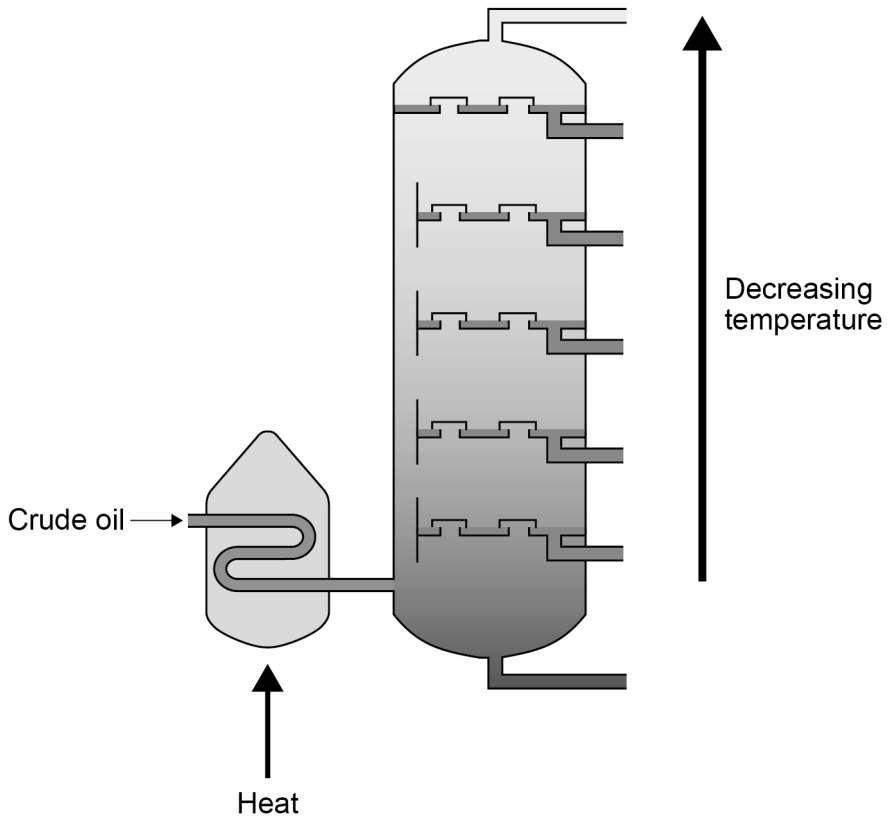
Question 1 continues on the next page

Do not write outside the box

0 1 . 3 Crude oil is a mixture of hydrocarbons.

Figure 1 represents industrial equipment used to separate crude oil into fractions.

Figure 1



Explain how crude oil is separated into fractions.

Use **Figure 1**.

[4 marks]

Do not write outside the box

0 1 . 4 The alkane molecule $C_{14}H_{30}$ can be cracked to produce smaller molecules.

Balance the equation for the reaction.

[1 mark]

$$C_{14}H_{30} \rightarrow C_8H_{18} + ____ \; C_3H_6$$

Question 1 continues on the next page

Do not write outside the box

Propene (C_3H_6) is an alkene.

0 1 . 5 Describe the test for alkenes.

Give the result.

[2 marks]

Test _____

_____

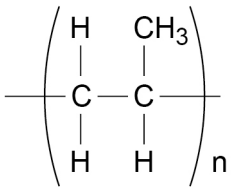
Result _____

_____

0 1 . 6 Poly(propene) is made from propene.

Figure 2 represents the repeating unit of poly(propene).

Figure 2



What type of substance is poly(propene)?

[1 mark]

_____

10